

Prathamesh Nehete

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EDUCATION

Northwestern University

M.S. in Data Science (AI Track)

Evanston, IL

Sep. 2025 – Sep. 2026

Arizona State University

B.S. in Computer Science; Business Minor

Tempe, AZ

Aug. 2021 – May 2025

WORK EXPERIENCE

Data Analyst

YYC Beeswax

Jan. 2025 – Aug. 2025

Tempe, AZ

- Analyzed sales and inventory data across 50+ SKUs to identify top-performing product lines and seasonal demand patterns, delivering insights that informed a 20% reduction in overstock carrying costs.
- Built interactive dashboards tracking revenue, customer acquisition, and order trends, enabling leadership to pinpoint high-margin craft categories and reallocate marketing spend accordingly.

Software Development Engineering Intern

Electro-Active Technologies

May 2024 – Aug. 2024

Knoxville, TN

- Analyzed real-time IoT sensor data from hydrogen conversion systems using Python, identifying process inefficiencies that informed a 15% improvement in conversion cycle optimization.
- Built automated data monitoring pipelines with REST APIs and AWS, reducing manual data quality checks by 40% and enabling stakeholders to make faster operational decisions.
- Translated cross-functional business requirements into data-driven dashboards and technical specs, presenting actionable insights to engineering leadership on system performance trends.

PROJECTS

Credit Risk Modelling and Fraud Detection: A Comprehensive Review | *IJERT*, Co-authored, Vol. 14, Aug. 2025

- Co-authored a peer-reviewed survey synthesizing classical and modern approaches to credit risk and fraud detection, comparing tree-based and deep learning methods, class-imbalance techniques (SMOTE, Tomek, cost-sensitive learning), evaluation metrics for skewed labels (PR-AUC, recall at fixed precision), and explainability frameworks (SHAP, LIME) used in production systems.

RA Flare Sentinel [\[link\]](#) | *Python, XGBoost, Optuna, SHAP, FastAPI, Kafka, DuckDB, React, MLflow, Docker*

- Built a 7-day rheumatoid-arthritis flare risk prediction pipeline (XGBoost, Optuna, SHAP) achieving 61.5% precision at 21.6% recall under a 10% clinician review budget, using patient-grouped walk-forward splits and isotonic calibration on longitudinal cohorts simulated from published RCT parameters (Wang et al., JMIR 2018; NCT02822521).
- Shipped a real-time inference service (FastAPI, Pydantic v2, Kafka, DuckDB, Docker Compose) with /predict, /shap-summary, /shap-row endpoints, MLflow-versioned artifacts, and a React + Recharts clinician dashboard surfacing cohort trajectories, calibration curves, per-patient SHAP, and HDBSCAN subgroup discovery.
- Hardened the codebase with GitHub Actions CI enforcing Ruff, strict Mypy, 80%+ Pytest coverage, Hypothesis property tests, and custom data-leakage and PHI-scan guardrails; authored model card, data card, threat model, and ADRs covering methodology, ethical scope, and reproducibility.

Credit Card Fraud Detection | *Python, XGBoost, SMOTE, SHAP, Scikit-learn, Matplotlib*

- Engineered end-to-end ML pipeline on highly imbalanced data (0.17% fraud rate), evaluating Logistic Regression, Random Forest, and XGBoost with SMOTE+Tomek resampling to achieve 95% precision at 87% recall.
- Delivered SHAP-based model explainability and false-positive cost analysis, enabling business stakeholders to quantify the cost-benefit tradeoff of fraud detection thresholds.

Airline Data Warehouse & Analytics | *PostgreSQL, MySQL, Python, ETL, Star Schema, Power BI*

- Engineered a star-schema data warehouse from 20M+ BTS flight records with automated ETL pipelines, enabling sub-second analytical queries and interactive Power BI dashboards for operational decision-making.
- Translated business KPIs into SQL-driven insights with dimensional modeling optimized for OLAP, supporting root-cause analysis of airline delays and performance benchmarking.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/TypeScript, C#, C/C++, SQL, Go, HTML/CSS, Bash

Cloud & DevOps: AWS (Lambda, EC2, S3, RDS, SageMaker), Azure, Docker, Terraform, Linux, CI/CD, Microservices

Data & ML: PostgreSQL, MySQL, Redis, Kafka, DuckDB, Pandas, Scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM, MLflow, SHAP, Optuna

Tools: FastAPI, Flask, ReactJS, REST APIs, Git, GitHub Actions, Jira, Agile, Grafana, Power BI, dbt

Certifications: Infosys Springboard — AI | Cloud Computing | Software Engineering